

Project Design Phase-II Technology Stack (Architecture & Stack)

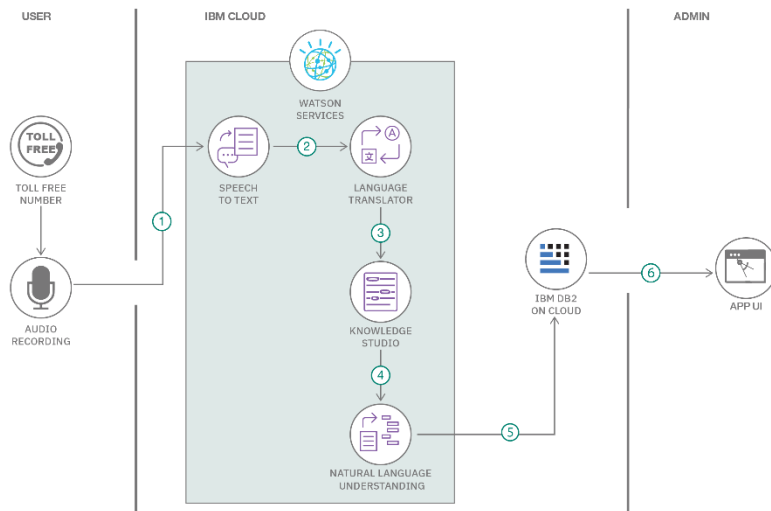
Date	15 June 2026
Team ID	Pulkit Khera
Project Name	OnlineComplaintRegistration
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1. 1	User Interface	How the user interacts with the application (Forms, Dashboards).	HTML, CSS, JavaScript, Bootstrap
2. 2	Application Logic-1	Core backend routing, form validation, and server logic.	Python, Flask Framework
3. 3	Application Logic-2	Database interactions and session management.	SQLAlchemy (Python ORM)
4. 4	Database	Relational database for storing users and tickets.	SQL (SQLite for dev / PostgreSQL for prod)
5. 5	Infrastructure	Application deployment and testing environment.	Local Server (Development)
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Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Core web framework utilized for rapid, modular development.	Flask (Python), Bootstrap (CSS)
2.	Security Implementations	Protection against SQL injection and unauthorized access; secure password storage.	SQLAlchemy, Werkzeug Security (Password Hashing)
3.	Scalable Architecture	Uses a Model-View-Controller (MVC) approach to separate database logic from UI routing.	Flask Blueprints / MVC Architecture
4.	Availability	The application will be accessible via standard web browsers globally once deployed.	Cloud Platform (e.g., Heroku, PythonAnywhere)
5.	Performance	Optimized database queries and lightweight frontend to ensure fast page loads.	SQL Indexing, Minified CSS/JS

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>